

“Oops, I Erred” — Collecting, Curating and Using Imperfect Data for Realistic Scenarios

Workshop

<https://example.com>

Organizers

Name	Affiliation	Email
Claas Voelcker	UT Austin	claas@cs.utexas.edu
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The corresponding organizers are **Claas Voelcker**, **Marcel Hussing**, and **Siddhant Agarwal**. They are supported by the **Oopsie-Data Team**, Arpit Bahety (UT Austin), Zhiyuan Zhou (UC Berkeley), Renhao Wang (UC Berkeley), Jiahui Chen (UT Austin), Carl Qi (UT Austin), Max Rudolph (UT Austin), and Maria Attarian, and Faculty Advisors Roberto Martín-Martín (UT Austin), Peter Stone (UT Austin), and Amy Zhang (UT Austin). As the full Oopsie Team is large, a core workshop team of three people was nominated to avoid diffusion of responsibility, with the full team providing support.

Organization Team Experience: The organizers have been involved with using imperfect data, especially failures, for improving robot policies [3, 2] and to learn world models [6, 9], reward models and failure taxonomies [12] and representations [1, 15, 18]. The members of the organization team have diverse organization experience. Claas Voelcker has organized Robot Learning 2023 and the Queer in AI 2021 and 2022 workshops at NeurIPS and is part of the organization team for the RL Conference 2026. Siddhant Agarwal has organized the RLBrew workshop at the RL Conferences 2024, 25, 26. Max Rudolph has organized the RLBrew workshop at the RL Conference 2026 and a workshop on interdisciplinary AI at the Trustworthy Autonomous Systems Conference. Zhiyuan Zhou has organized Eval&Deploy Workshop at CoRL 2025. Arpit Bahety is organizing the workshop on Rethinking Safety at RSS 2026. Advising faculty members have organized a large variety of workshops, conferences, and other events.

Description

Recent advances in embodied artificial intelligence have been driven by the development and availability of large-scale datasets. In robot manipulation, datasets such as DROID [8] and the Open Cross-Embodiment Dataset (OXE) [14] have enabled significant progress by focusing on successful task execution. In robot manipulation, DROID [8] and the Open Cross-Embodiment Dataset (OXE) [14] enabled rapid progress by aggregating successful task executions; analogous efforts in autonomous driving and embodied navigation, such as the Waymo Open Motion Dataset [7] and Room-to-Room [10], did the same for nominal driving and successful navigation routes. Training on curated successes collected in controlled settings has arguable lead to models which master a diversity of tasks but degrade under the uncertainty of real deployment.

We use *imperfect data* to refer to robot experience that differ from these clean, expertly curated demonstrations: execution failures and near-misses, suboptimal or noisy demonstrations, out-of-distribution events, and unexpected human interactions during deployment. Precisely this data which could close the gap to real deployment is often fragmented, undocumented, and largely inaccessible across robotics labs.

This raises a fundamental question: can imperfect data help robots succeed in realistic, imperfect scenarios, and what challenges arise in collecting, curating, and using it? Execution failures, for in-

stance, are critical for understanding robustness, improving policies, and ensuring safe deployment, yet it remains unclear how to label failures or even which kinds of failure matter most for progress. These questions cannot be answered by individual labs alone; they require a community-wide effort to debate what data we need and how to share it. The largest robotics datasets increasingly emerge from multi-lab collaborations, but traditional academic structures such as conference publications leave little room for the coordination such efforts demand, and for proper incentive structures and rewards for data work. This workshop asks how a shared infrastructure for collecting, representing, curating, and publishing imperfect data can be built across academia and industry.

The workshop’s outcomes will be documented in a workshop white-paper that collects input from both submitted works and workshop discussions. Our main goal is to summarize community opinions on dataset design and the infrastructure that enables sharing heterogeneous and imperfect data, how imperfect data can be categorized and made legible to the community, and finally downstream applications that should be prioritized by dataset providers and contributors.

Core Challenges / Research Questions

1. **How can we create robotic datasets of imperfect scenarios at scale?** What barriers to data sharing, standardization, and infrastructure must be overcome?
2. **How can we make sense of such data?** How should imperfect data be classified, annotated, and formatted for broad usability?
3. **How can imperfect data improve downstream robot learning?** Which types of imperfect data actually help, and is some data simply too messy to be worth keeping?

Invited Speakers

All confirmed speakers have only committed to presenting roles in at max two workshops.

Andrea Bajcsy & Student (Keynote)

IN-PERSON

Carnegie Mellon University

`speaker.x@iix.edu`

Andrea Bajcsy is an Assistant Professor in CMU’s Robotics Institute, where she leads the Interactive and Trustworthy Robotics Lab. Her work spans robotics, machine learning, and human-AI interaction, with a sustained focus on agents that operate reliably under imperfect models and data [4, 13, 16]. She will co-present the keynote with a student author of her ICML 2026 spotlight position paper, “Good Embodied Reward Models Need Bad Behavior Data” [17], which argues that robotics must invest in failed and hazardous behavior data rather than training only on clean demonstrations, which directly establishes a position to this workshop’s core question.

Pepijn Kooijmans or team member (Panelist)

IN-PERSON

HuggingFace

`speaker.x@iix.edu`

Pepijn Kooijmans is a core member of the Hugging Face LeRobot team and co-author of the library’s ICLR 2026 publication [5], working across the LeRobotDataset format, hardware integration, and policies.¹ The team built the standardized LeRobotDataset format and Hub tooling specifically to tackle robotics’ data-fragmentation problem and is explicitly community-driven, giving him a hands-on view of how robot data is collected, curated, and shared at scale.

Karl Pertsch (Panelist)

IN-PERSON

Physical Intelligence

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¹<https://github.com/huggingface/lerobot>

Karl Pertsch is a member of the technical staff as Physical Intelligence. He was a postdoc at UC Berkeley and Stanford University, and received his PhD from the University of Southern California (USC). He lead the DROID dataset effort, a multi-lab dataset generation and curation effort that forms an important pillar of modern manipulation model training.

Yang Gao (Panelist)

TENTATIVE

Tsinghua University

speaker.x@iix.edu

Yang Gao is an Assistant Professor at Tsinghua University working at the intersection of computer vision and robotics. He has worked on data scaling laws for imitation learning [11], and he has further studied robot data augmentation [19] and manipulation learning from sparse failure guidance [20]. His research speaks directly to how imperfect, limited, or visually varied data can be processed and used to train more capable policies.

Julie Shah (Panelist)

REQUESTED

MIT and Lila Sciences

speaker.x@iix.edu

Julie Shah is a Professor at MIT and Head of the Department of Aeronautics and Astronautics, and recently became Chief Robotics Officer at Lila Sciences. Her work on human-robot collaboration brings the perspective of what data is needed when robots learn from, coordinate with, and adapt to people.

Dorsa Sadigh (Panelist)

REQUESTED

Stanford University

speaker.x@iix.edu

Dorsa Sadigh is an Associate Professor of Computer Science at Stanford, working at the intersection of robot learning and human-robot interaction. Her research on learning from imperfect and suboptimal demonstrations, from demonstrators with mismatched dynamics, and from human feedback offers a direct perspective on how robots can exploit data that is noisy or shaped by human interaction.

Proposed Format and Interactivity

Tentative schedule (assuming morning session)

Time	Program	Speaker
8:30–8:45	Opening remarks	Organizers
8:45–9:30	Keynote Address	Andrea Bajcsy & student
9:30–10:30	Community Collaboration Session	<i>Community</i>
10:30–11:30	Coffee Break & Poster Session	<i>Community</i>
11:30–12:00	Panel “What is good data for physical AI? How do we obtain it, process it, and use it?”	Panelists (Pepjin Kooijmans, Yang Gao, Karl Pertsch, Julie Shah, Dorsa Sadigh)
12:15–12:30	Closing Remarks	Organizers

Keynote & Community Collaboration Session. The keynote introduces the topic and sets up the discussion. The core of the workshop is a breakout session in which participants will be split up into smaller groups to discuss questions along the major axes of the topic: “What incentive structures sustain long-term data-sharing?”, “Which kinds of imperfect data are useful for participants’ research?”, and “How should infrastructure for heterogeneous imperfect data be designed?” Junior community members facilitate the discussions and take notes, feeding directly into the workshop white-paper. Claas Voelcker has helped organize a similar format successfully at a Queer in AI governance workshop at NeurIPS.

Poster Session. The poster session offers a further channel for participants to share their perspective on the workshop topics (see contributions below).

Panel. To present the range of perspectives on imperfect data, we close with a panel on “What is good data for physical AI? How do we obtain it, process it, and use it?”, bringing together industry, start-up, and academic voices spanning data hosting, learning from data, and human-robot interaction.

Awards. The panel is scheduled immediately before the dataset and paper awards. Some awards follow from reviewer feedback; others are decided live. For the live categories — e.g., “funniest failure” or “most unlikely failure” — participants submit examples beforehand, a pre-selected subset is shown live, and the audience votes on the most representative submission, transitioning directly into the awards ceremony. We believe involving the community directly in award selection, as opposed to decisions made by intransparent committees, will increase the engagement of attendees.

Contributed Papers Plan

We solicit two kinds of contributions: papers and datasets. The organizing team currently coordinates the Oopsie-Data effort, which collects a large database of robot manipulation failures and makes them accessible to the community. Because incentives for contributing to large-scale data efforts in academia remain limited — dedicated dataset venues are rare outside the NeurIPS benchmark track — establishing a dataset track is a concrete first step toward changing this.

Papers may be short (4 pages) or long (8 pages) and cover any aspect of collecting, curating, or using imperfect data, including position papers arguing for or against the usefulness of particular forms of imperfect data. All submissions are reviewed double-blind on OpenReview by a program committee distinct from the organizers, and selected for correctness, coherence, and potential to spark discussion, prioritizing experimental and early-stage work. Following CoRL policy, we will not accept work already accepted to the main conference.

Datasets are solicited as first-class contributions. Each must be accompanied by a short white-paper (max 4 pages) describing the data’s origin, use-case, and scope, with documentation following the Oopsie formatting standards; review focuses on documentation quality, community usability, and topical fit. Contributors present via a poster, video, or an artifact such as a trained model demonstrated on a real system.

To reward the often-invisible labor of dataset creation and curation, we extend the award mechanisms normally reserved for papers to datasets: participants and panelists vote for their favorite data contributions during the poster session, with winners announced at the end of the panel.

Plans to Encourage Participation and Diversity

Our speakers are diverse across geography (North America, Europe, East Asia), gender, and sector (industry and academia), and our keynote speaker will co-present with one of her students to give visibility to a junior researcher. The collaboration-session round tables will likewise be facilitated by graduate students.

To reach beyond the invited list, we will lean on community networks. Two organizers (Claas and Roberto) are organizing members of Queer in AI and Queer in Robotics, and we will promote the workshop through Queer in AI’s world-wide channels as well as those of Women in Machine Learning, Black in AI, Muslim in AI, and Disability in AI.

Logistics and Resources

Estimated audience size: 100

Physical requirements: We require the ability to rearrange the seating in the room to enable non front-facing discussion groups. Preferably, this would be facilitated through several large round

tables. We require poster boards for approximately 50 small posters. Finally, if possible, we would request several screens to display dataset contributions in the form of videos.

Special requirements: n/a

Commitments

- ✓ At least one organizer will attend CoRL 2026 in person.
- ✓ Organizers will attend the CoRL 2026 workshop organizers meeting.
- ✓ Organizers will submit a post-workshop artifact.
- ✓ Willingness to merge with related proposals if requested.

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